

MATHEMATICS EXTENSION 1

Time allowed: Two hours (plus 5 minutes reading) **Exam date:** 13th August 2003

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the right margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- Each question will be collected separately.
- Start each question in a new answer booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each answer booklet:

Checklist:

- SGS Examination Booklets required — seven 4-page booklets per boy.
- Candidature: 120 boys.

QUESTION ONE (Start a new answer booklet)

Marks

(a) Solve the inequation $\frac{1}{x-3} < 3$.

2

(b) Evaluate $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$, giving your answer in exact form.

2

(c) Differentiate with respect to x :

(i) $y = \tan^{-1} 2x$

1

(ii) $y = \log_e \cos x$

2

(d) Find, correct to the nearest degree, the acute angle between the straight lines $y = 3$ and $y = -\frac{5}{3}x + 2$.

2

(e) Let α , β and γ be the roots of $2x^3 - x^2 + 3x - 2 = 0$. Find the value of

3

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}.$$

QUESTION TWO (Start a new answer booklet)

Marks

(a) Use the substitution $u = 1 + \tan x$ to evaluate $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1 + \tan x}} dx$.

3

(b) Find the term independent of x in the expansion of $\left(x^2 - \frac{3}{x^2}\right)^6$.

3

(c) Using the t -substitutions, or otherwise, prove the identity

3

$$\frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta} = \tan^2 \theta.$$

(d) An object, always spherical in shape, is increasing in volume at a constant rate of $8 \text{ m}^3/\text{min}$.(i) Find the rate at which the radius is increasing when the radius is 4 metres.
(Note: You may assume the volume formula $V = \frac{4}{3}\pi r^3$).

2

(ii) Find the rate at which the surface area is increasing when the radius is 4 metres.
(Note: You may assume the surface area formula $S = 4\pi r^2$).

1

QUESTION THREE (Start a new answer booklet)

- (a) Consider the function $f(x) = 3\sin^{-1}(x + 1)$. Marks
- (i) Write down the domain and the range of $f(x)$. 2
- (ii) Sketch $y = f(x)$, giving the coordinates of its endpoints and any intercepts with the coordinate axes. 2
- (b) A particle moves according to the equation $v^2 = 2x(6 - x)$.
- (i) Show that the particle moves in the interval $0 \leq x \leq 6$. 1
- (ii) Write down the centre of the motion. 1
- (iii) Find the maximum speed of the particle. 1
- (iv) Find the acceleration function. 1
- (c) The expression $\left(2 + \frac{x}{3}\right)^n$ is expanded. The ratio of the coefficients of the terms in x^6 and x^7 is $7 : 8$. Find the value of n . 4

QUESTION FOUR (Start a new answer booklet)

- (a) The polynomial $2x^3 + ax^2 + bx + 6$ has $x - 1$ as a factor and leaves a remainder of -12 when divided by $x + 2$. Find the values of a and b . Marks 4
- (b) Given that the equation $x^3 + px^2 + qx + r = 0$ has a triple root, use the sums and products of roots to show that $pq = 9r$. (Hint: Let the roots be α, α and α). 4
- (c) (i) Show that the coefficient of x^5 in the expansion of $(1 + x)^4(1 + x)^4$ is given by 3
- $${}^4C_0 \times {}^4C_1 + {}^4C_1 \times {}^4C_2 + {}^4C_2 \times {}^4C_3 + {}^4C_3 \times {}^4C_4.$$
- (ii) Hence, by equating the coefficients of x^5 on both sides of the identity 1
- $$(1 + x)^4(1 + x)^4 = (1 + x)^8,$$
- prove that ${}^4C_0 \times {}^4C_1 + {}^4C_1 \times {}^4C_2 + {}^4C_2 \times {}^4C_3 + {}^4C_3 \times {}^4C_4 = \frac{8!}{3! \times 5!}.$

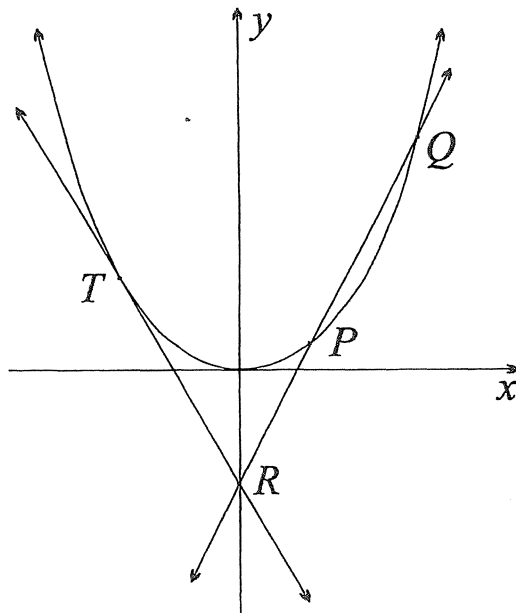
QUESTION FIVE (Start a new answer booklet)

- (a) The temperature of a body is changing at the rate $\frac{dT}{dt} = -k(T - 20)$, where T is the temperature at time t minutes and k is a positive constant.

The temperature of the surrounding environment is 20°C . The initial temperature of the body is 36°C and it falls to 35°C in 5 minutes:

- (i) Show that $T = 20 + Ae^{-kt}$ is a solution of $\frac{dT}{dt} = -k(T - 20)$, where A is a constant. Marks 1
- (ii) Prove that $A = 16$ and $k = -\frac{1}{5}\log_e \frac{15}{16}$. 3
- (iii) Find how long, correct to the nearest minute, it will take the temperature to fall to 27°C . 2
- (iv) Explain why the body will never reach a temperature that is one half of its initial temperature. 1

(b)



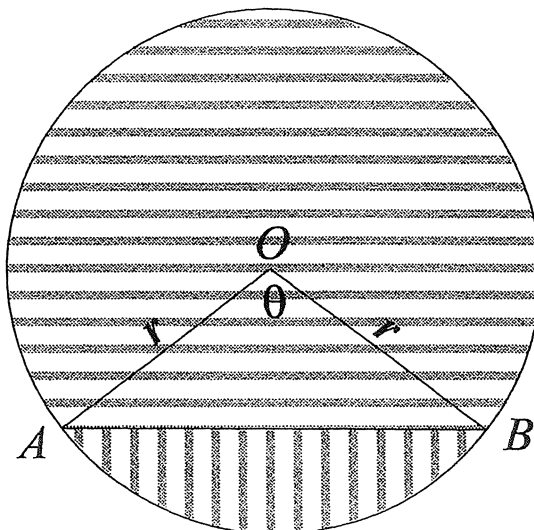
The diagram above shows the parabola $x^2 = 4ay$. The points $T(2at, at^2)$, $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola.

You may assume that the chord PQ has equation $y - \frac{1}{2}(p + q)x + apq = 0$.

- (i) Prove that the equation of the tangent to the parabola at the point $T(2at, at^2)$ is $y - tx + at^2 = 0$. 2
- (ii) Let the tangent at T intersect the axis of the parabola at the point R . Find the coordinates of R . 1
- (iii) Given that the chord PQ also passes through R , show that the parameters p , t and q form a geometric sequence. 2

QUESTION SIX (Start a new answer booklet)

(a)



In the diagram above, the chord AB subtends an angle of θ radians at the centre O of the circle with radius r .

- (i) Show that the ratio of the areas of the two segments is

Marks

2

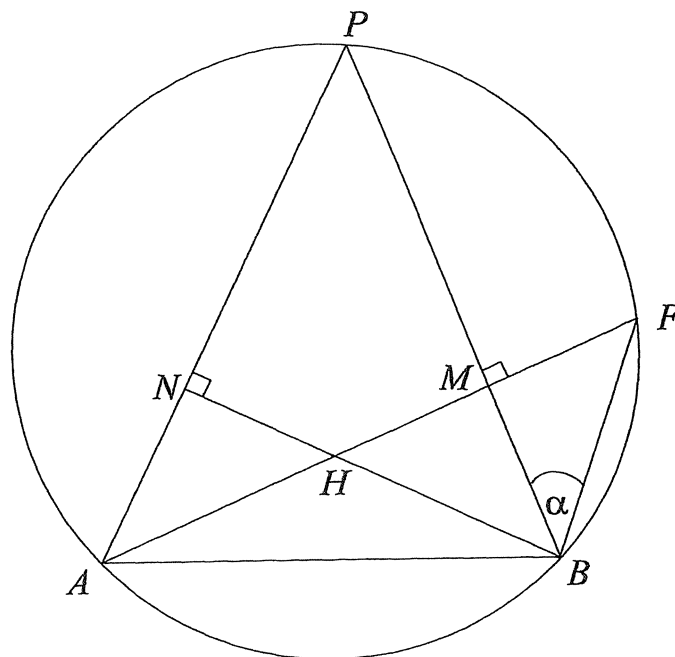
$$\frac{\text{area of major segment}}{\text{area of minor segment}} = \frac{2\pi - \theta + \sin \theta}{\theta - \sin \theta}.$$

- (ii) Now suppose that

$$\frac{\text{area of major segment}}{\text{area of minor segment}} = \frac{\pi - 1}{1}.$$

- (α) Prove that $\theta - 2 - \sin \theta = 0$. **1**
- (β) Show that the equation $\theta - 2 - \sin \theta = 0$ has a root between $\theta = 2$ and $\theta = 3$. **1**
- (γ) Taking $\theta = 2.5$ as the first approximation, use Newton's method to find a second approximation to the root. Give your answer correct to two decimal places. **1**
- (δ) Determine whether the second approximation of θ yields a smaller value of $|\theta - 2 - \sin \theta|$ than the first approximation. **1**

(b)



In the diagram above, ABP is a triangle inscribed in a circle.

The altitudes BN and AM of the triangle intersect at H .

The altitude AM is produced to meet the circumference of the circle at F .

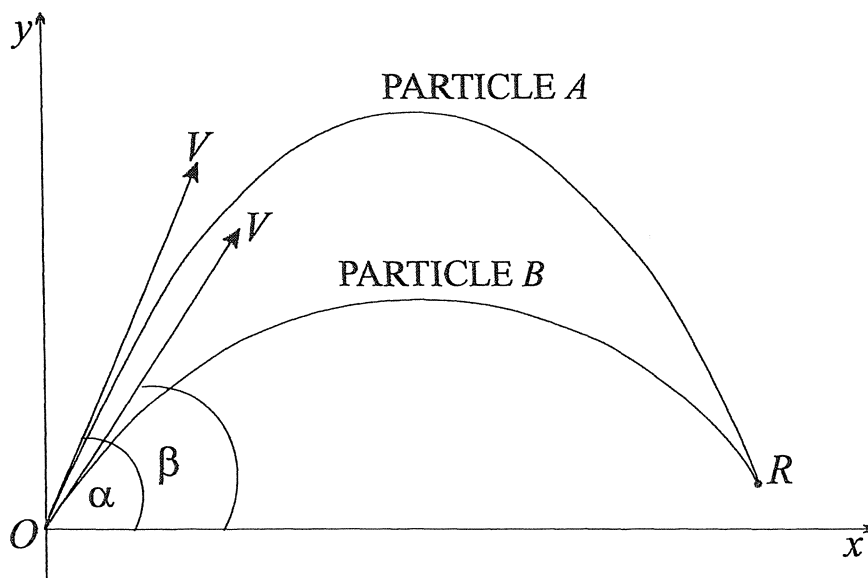
Copy the diagram into your examination booklet.

Let $\angle PBF = \alpha$.

- | | |
|---|---|
| (i) Why is $\angle PAF = \alpha$? | 1 |
| (ii) Why are the points A, N, M , and B concyclic? | 1 |
| (iii) Why is $\angle NBM = \alpha$? | 1 |
| (iv) Show that M bisects HF . | 2 |
| (v) If AB is a fixed chord of the circle and P moves on the major arc AB , show that α is independent of the position of P . | 1 |

QUESTION SEVEN (Start a new answer booklet)

(a)



The diagram above shows two particles *A* and *B* projected from the origin.

Particle *A* is projected with initial velocity V m/s at an angle α .

Particle *B* is projected T seconds later with the same initial velocity V m/s but at an angle of β .

The particles collide at the point *R*.

(i) You may assume that the equations of the paths of *A* and *B* are:

Marks

3

$$\text{For } A: y = -\frac{gx^2}{2V^2} \sec^2 \alpha + x \tan \alpha$$

$$\text{For } B: y = -\frac{gx^2}{2V^2} \sec^2 \beta + x \tan \beta$$

Show that the x -coordinate of the point *R* of collision is

$$x = \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)}.$$

(ii) You may assume that the equation of the horizontal displacement of *A* is

$$x = Vt \cos \alpha.$$

(α) Write down the equation for the horizontal displacement of *B*. (Remember that *B* is projected T seconds after *A*).

1

(β) Show that the difference T in the times of projection is

2

$$T = \frac{2V(\cos \beta - \cos \alpha)}{g \sin(\alpha + \beta)}.$$

- (b) (i) Prove by mathematical induction that for all positive integers n ,

4

$$\sin(n\pi + x) = (-1)^n \sin x.$$

- (ii) Let $S = \sin(\pi + x) + \sin(2\pi + x) + \sin(3\pi + x) + \cdots + \sin(n\pi + x)$, for $0 < x < \frac{\pi}{2}$ and for all positive integers n . Show that

2

$$-1 < S \leq 0.$$

GJ

QUESTION ONE

(a) $\frac{1}{x-3} < 3, x \neq 3$

$$\frac{1}{x-3} \times (x-3)^2 < 3(x-3)^2$$

$$x-3 < 3(x-3)^2 \quad \checkmark$$

$$3(x-3)^2 - (x-3) > 0$$

$$(x-3)(3(x-3)-1) > 0$$

$$(x-3)(3x-10) > 0$$

$$x < 3 \text{ or } x > \frac{10}{3} \quad \checkmark$$

(b) $\int_0^3 \frac{dx}{\sqrt{9-x^2}} = \left[\sin^{-1} \frac{x}{3} \right]_0^3 \quad \checkmark$
 $= \sin^{-1} 1 - \sin^{-1} 0$
 $= \frac{\pi}{2} \quad \checkmark$

(c) (i) $y = \tan^{-1} 2x$

$$\frac{dy}{dx} = \frac{2}{1+4x^2} \quad \checkmark$$

(ii) $y = \log_e \cos x$

$$\frac{dy}{dx} = -\frac{\sin x}{\cos x} \quad \checkmark \text{ for } -\sin x \quad \checkmark \text{ for quotient}$$

(d) $\tan \theta = \left| -\frac{5}{3} \right| \quad \checkmark$
 $\theta \doteq 59^\circ \quad \checkmark$

(e) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \quad \checkmark$
 $= \frac{3}{2} \div 1$
 $= \frac{3}{2} \quad \checkmark \checkmark \text{ any correct method}$

QUESTION TWO

(a) $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1+\tan x}} dx = \int_1^2 \frac{du}{u^{\frac{1}{2}}} \quad \checkmark$
 $= \int_1^2 u^{-\frac{1}{2}} du$
 $= \left[2u^{\frac{1}{2}} \right]_1^2 \quad \checkmark$
 $= 2\sqrt{2} - 2 \quad \checkmark$

Let $u = 1 + \tan x$

$du = \sec^2 x dx$

When $x = 0$, $u = 1$,

When $x = \frac{\pi}{4}$, $u = 2$.

(b) General term $= {}^6C_r (x^2)^{6-r} (-1)^r (3x^{-2})^r$
 $= {}^6C_r (x)^{12-2r} (-1)^r (3)^r (x)^{-2r}$
 $= {}^6C_r (-1)^r (3)^r (x)^{12-4r} \quad \checkmark$

Let $12 - 4r = 0$

$r = 3 \quad \checkmark$

Term independent of $x = {}^6C_3 (-1)^3 (3)^3$
 $= -540 \quad \checkmark$

(c) $LHS = \frac{\tan 2\theta - \tan \theta}{\tan 2\theta + \cot \theta}$

Let $t = \tan \theta$

$$LHS = \left(\frac{2t}{1-t^2} - t \right) \div \left(\frac{2t}{1-t^2} + \frac{1}{t} \right) \quad \checkmark$$

$$= \frac{2t-t+t^3}{1-t^2} \times \frac{t(1-t^2)}{2t^2+1-t^2}$$

$$= \frac{t(1+t^2)}{1-t^2} \times \frac{t(1-t^2)}{t^2+1}$$

$\checkmark$ correct method of simplification of the algebraic fractions

$$= t^2 \quad \checkmark$$

$$= \tan^2 \theta$$

$$= RHS$$

(d) (i) $V = \frac{4}{3}\pi r^3$

$$\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt} \quad \checkmark$$

$$8 = 64\pi \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{8\pi} \text{ m/min} \quad \checkmark$$

(ii) $S = 4\pi r^2$

$$\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

$$= 8\pi r \times \frac{1}{8\pi}$$

$$= 4 \text{ m}^2/\text{min} \quad \checkmark$$

SECTION THREE

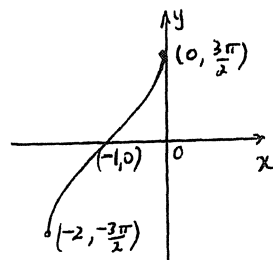
b) (i) $f(x) = 3 \sin^{-1}(x+1)$

Domain: $-1 \leq x+1 \leq 1$

$-2 \leq x \leq 0$ ☒

Range: $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$ ☒

(ii)



☒ Shape

☒ Labels

b) (i) $v^2 = 2x(6-x)$

$2x(6-x) \geq 0$

$0 \leq x \leq 6$ ☒

(ii) $x = 3$ ☒

(iii) Maximum speed when $x = 3$.

$v^2 = 6 \times 3$

$|v| = 3\sqrt{2}$ ☒

(iv) $v^2 = 2x(6-x)$

$\frac{1}{2}v^2 = 6x - x^2$

$\frac{d}{dx}(\frac{1}{2}v^2) = 6 - 2x$

$\ddot{x} = 6 - 2x$ ☒

c) Given $\left(2 + \frac{x}{3}\right)^n$:

term in $x^6 = {}^nC_6 \times 2^{n-6} \times \left(\frac{x}{3}\right)^6$

term in $x^7 = {}^nC_7 \times 2^{n-7} \times \left(\frac{x}{3}\right)^7$

☒ 1 mark for both answers

$$\begin{aligned} \text{Ratio of coefficients} &= \frac{\frac{n!}{6!(n-6)!} \times 2^{n-6} \times \left(\frac{1}{3}\right)^6}{\frac{n!}{7!(n-7)!} \times 2^{n-7} \times \left(\frac{1}{3}\right)^7} \\ &= \frac{n!}{6!(n-6)!} \times \frac{7!(n-7)!}{n!} \times 3 \times 2 \quad \boxed{\checkmark} \\ &= \frac{42}{n-6} \quad \boxed{\checkmark} \\ \text{so } \frac{7}{8} &= \frac{42}{n-6} \\ n-6 &= 48 \\ n &= 54. \quad \boxed{\checkmark} \end{aligned}$$

QUESTION FOUR

(a) Let $P(x) = 2x^3 + ax^2 + bx + 6$

$$P(1) = 2 + a + b + 6$$

$$0 = a + b + 8$$

$$a + b = -8$$

$$\dots (1) \quad \boxed{\checkmark \text{ for any correct form}}$$

$$P(-2) = -16 + 4a - 2b + 6$$

$$-12 = 4a - 2b - 10$$

$$4a - 2b = -2$$

$$2a - b = -1$$

$$\dots (2) \quad \boxed{\checkmark \text{ for any correct form}}$$

$$(1) + (2) \quad 3a = -9$$

$$a = -3 \quad \boxed{\checkmark}$$

$$b = -5 \quad \boxed{\checkmark}$$

(b) $x^3 + px^2 + qx + r = 0$

$$3\alpha = -p$$

$$\dots (1) \quad \boxed{\checkmark}$$

$$3\alpha^2 = q$$

$$\dots (2) \quad \boxed{\checkmark}$$

$$\alpha^3 = -r$$

$$\dots (3) \quad \boxed{\checkmark}$$

$$(1) \times (2) \quad 9\alpha^3 = -pq$$

$$-9r = -pq$$

$$pq = 9r \quad \boxed{\checkmark}$$

(c) (i) $(1+x)^4(1+x)^4 = ({}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4)$

$$\times ({}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4) \quad \boxed{\checkmark}$$

$$\text{Term in } x^5 = {}^4C_1x \times {}^4C_4x^4 + {}^4C_2x^2 \times {}^4C_3x^3 + {}^4C_3x^3 \times {}^4C_2x^2 + {}^4C_4x^4 \times {}^4C_1x \quad \boxed{\checkmark}$$

$$\text{Coefficient} = {}^4C_1 \times {}^4C_4 + {}^4C_2 \times {}^4C_3 + {}^4C_3 \times {}^4C_2 + {}^4C_4 \times {}^4C_1$$

$$= {}^4C_0 \times {}^4C_1 + {}^4C_1 \times {}^4C_2 + {}^4C_2 \times {}^4C_3 + {}^4C_3 \times {}^4C_4, \text{ by symmetry.} \quad \boxed{\checkmark}$$

(ii) Coefficient of x^5 in $(1+x)^8 = {}^8C_5$

$$= \frac{8!}{3! \times 5!} \quad \boxed{\checkmark}$$

$$\text{Now } (1+x)^4(1+x)^4 = (1+x)^8,$$

$$\text{so } {}^4C_0 \times {}^4C_1 + {}^4C_1 \times {}^4C_2 + {}^4C_2 \times {}^4C_3 + {}^4C_3 \times {}^4C_4 = \frac{8!}{3! \times 5!}.$$

QUESTION FIVE

- (a) (i) Given
- $T = 20 + Ae^{-kt}$

$$\frac{dT}{dt} = -kAe^{-kt}$$

$$= -k(T - 20). \quad \boxed{\checkmark}$$

So $T = 20 + Ae^{-kt}$ is a solution.

- (ii) When
- $t = 0$
- ,
- $T = 36$

$$\text{so } 36 = 20 + Ae^0$$

$$A = 16. \quad \boxed{\checkmark}$$

When $t = 5$, $T = 35$

$$\text{so } 35 = 20 + 16e^{-5k}$$

$$15 = 16e^{-5k}$$

$$e^{-5k} = \frac{15}{16} \quad \boxed{\checkmark}$$

$$-5k = \log_e \frac{15}{16}$$

$$k = -\frac{1}{5} \log_e \frac{15}{16}. \quad \boxed{\checkmark}$$

- (iii) When
- $T = 27$
- ,

$$27 = 20 + 16e^{-kt}$$

$$e^{-kt} = \frac{7}{16} \quad \boxed{\checkmark}$$

$$t = \frac{\log_e \frac{7}{16}}{-k}$$

$$= 64.045 \dots$$

It will take 64 minutes. $\boxed{\checkmark}$

- (iv) As
- $t \rightarrow \infty$
- ,
- $T \rightarrow 20$
- from above.

The temperature does not drop below 20°C and so will never reach 18°C . $\boxed{\checkmark}$

$$(b) (i) \quad y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$\text{At } T, \quad \frac{dy}{dx} = \frac{2at}{2a}$$

$$= t. \quad \boxed{\checkmark}$$

$$\text{Now } y - at^2 = t(x - 2at)$$

$$y - at^2 = tx - 2at^2$$

$$\text{so } y - tx + at^2 = 0. \quad \boxed{\checkmark}$$

- (ii) Let
- $x = 0$

$$\text{so } y = -at^2$$

 R is the point $(0, -at^2)$. $\boxed{\checkmark}$

- (iii)
- R
- lies on
- PQ
- .

$$y - \frac{1}{2}(p+q)x + apq = 0$$

$$-at^2 + apq = 0 \quad \boxed{\checkmark}$$

$$t^2 = pq, \quad a \neq 0$$

$$\frac{t}{p} = \frac{q}{t}$$

So p , t , and q form a geometric sequence. $\boxed{\checkmark}$

QUESTION SIX

(a) (i) Area of minor segment = $\frac{1}{2}r^2(\theta - \sin \theta)$

Area of major segment = $\pi r^2 - \frac{1}{2}r^2(\theta - \sin \theta)$

$$\begin{aligned} \text{Ratio of areas} &= \frac{\pi r^2 - \frac{1}{2}r^2(\theta - \sin \theta)}{\frac{1}{2}r^2(\theta - \sin \theta)} \quad \checkmark \\ &= \frac{2\pi - \theta + \sin \theta}{\theta - \sin \theta} \quad \checkmark \end{aligned}$$

(ii) (α) $\frac{2\pi - \theta + \sin \theta}{\theta - \sin \theta} = \frac{\pi - 1}{1}$

$$\begin{aligned} \pi\theta - \pi\sin\theta - \theta + \sin\theta &= 2\pi - \theta + \sin\theta \\ \theta - 2 - \sin\theta &= 0 \quad \checkmark \end{aligned}$$

(β) Let $f(\theta) = \theta - 2 - \sin \theta$

$$f(2) = -\sin 2$$

$$\doteq -0.909$$

$$< 0$$

$$f(3) = 1 - \sin 3$$

$$\doteq 0.859$$

$$> 0.$$

So the root lies between $\theta = 2$ and $\theta = 3$, $\checkmark$

(γ) $f(\theta) = \theta - 2 - \sin \theta$

$$f'(\theta) = 1 - \cos \theta.$$

Let θ_0 be the first approximation.

$$\theta_1 = \theta_0 - \frac{\theta_0 - 2 - \sin \theta_0}{1 - \cos \theta_0}$$

$$\theta_1 = 2.5 - \frac{2.5 - 2 - \sin 2.5}{1 - \cos 2.5} \quad \checkmark$$

$$\doteq 2.55$$

(δ) When $\theta = 2.5$,

$$|\theta - 2 - \sin \theta| \doteq 0.09847.$$

(ε) When $\theta = 2.55$,

$$|\theta - 2 - \sin \theta| \doteq 0.00768. \text{ So } \theta = 2.55 \text{ yields a smaller value. } \checkmark$$

(b) (i) $\angle PAF = \angle PBF$ angles at circumference standing on the same arc $\checkmark$
 $\angle PAF = \alpha.$

(ii) $\angle ANB = \angle AMB$ (both given as rightangles).

These lie on the same interval AB and so A,N,M and B are concyclic. $\checkmark$

(iii) $\angle NBM = \angle MAN$ (angles standing on the same arc of circle $ANMB$) $\checkmark$
 $\angle NBM = \alpha.$

(iv) $\triangle BHM \equiv \triangle BFM$ (AAS test) $\checkmark$

$$HM = MF \text{ (matching sides of congruent triangles) } \checkmark$$

(v) $\angle APB$ stands on fixed chord AB and its size is independent of the position of P (angles at circumference standing on the same chord). So α is independent of the position of P . $\checkmark$

QUESTION SEVEN

(a) (i) For A: $y = -\frac{gx^2}{2V^2} \sec^2 \alpha + x \tan \alpha \dots (1)$

For B: $y = -\frac{gx^2}{2V^2} \sec^2 \beta + x \tan \beta \dots (2)$

At R the coordinates are identical, so substitute (1) in (2).

$$-\frac{gx^2}{2V^2} \sec^2 \alpha + x \tan \alpha = -\frac{gx^2}{2V^2} \sec^2 \beta + x \tan \beta \quad \checkmark$$

$$\frac{gx^2}{2V^2} (\sec^2 \alpha - \sec^2 \beta) = x (\tan \alpha - \tan \beta)$$

$$\frac{gx}{2V^2} (\tan^2 \alpha - \tan^2 \beta) = (\tan \alpha - \tan \beta), \quad x \neq 0 \quad \checkmark$$

$$\frac{gx}{2V^2} = \frac{(\tan \alpha - \tan \beta)}{(\tan^2 \alpha - \tan^2 \beta)}$$

$$x = \frac{2V^2}{g} \times \frac{1}{\tan \alpha + \tan \beta}, \quad \tan \alpha \neq \tan \beta$$

$$= \frac{2V^2}{g} \times \frac{\cos \alpha \cos \beta}{\sin \alpha \cos \beta + \cos \alpha \sin \beta}$$

$$= \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)} \quad \checkmark$$

(ii) (α) $x = V(t - T) \cos \beta. \quad \checkmark$

(β) When A is at R:

$$Vt \cos \alpha = \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)}$$

$$t = \frac{2V \cos \beta}{g \sin(\alpha + \beta)} \dots (3) \quad \checkmark$$

When B is at R:

$$V(t - T) \cos \beta = \frac{2V^2 \cos \alpha \cos \beta}{g \sin(\alpha + \beta)}$$

$$t - T = \frac{2V \cos \alpha}{g \sin(\alpha + \beta)}$$

$$T = t - \frac{2V \cos \alpha}{g \sin(\alpha + \beta)}$$

$$= \frac{2V \cos \beta}{g \sin(\alpha + \beta)} - \frac{2V \cos \alpha}{g \sin(\alpha + \beta)}, \text{ from (3)}$$

$$= \frac{2V(\cos \beta - \cos \alpha)}{g \sin(\alpha + \beta)} \quad \checkmark$$

(b) (i) Prove by mathematical induction the proposition that for all positive integers

$$\sin(n\pi + x) = (-1)^n \sin x, \text{ for } 0 < x < \frac{\pi}{2}.$$

A. When $n = 1$,

$$LHS = \sin(\pi + x)$$

$$= -\sin x$$

$$= RHS.$$

The proposition is true for $n = 1. \quad \checkmark$

B. Assume the proposition is true for some positive integer k so that $\sin(k\pi + x) = (-1)^k \sin x \dots (*)$

We are required to prove the proposition true for $n = k + 1$.

That is, $\sin[(k + 1)\pi + x] = (-1)^{k+1} \sin x. \quad \checkmark$

Now

$$LHS = \sin[(k + 1)\pi + x]$$

$$= \sin[\pi + (k\pi + x)] \quad \checkmark$$

$$= \sin \pi \cos(k\pi + x) + \cos \pi \sin(k\pi + x)$$

$$= -1 \times \sin(k\pi + x)$$

$$= -1 \times (-1)^k \sin x, \text{ from } (*) \quad \checkmark$$

$$= (-1)^{k+1} \sin x$$

$$= RHS$$

It follows from A and B by mathematical induction that for all positive integers n , $\sin(n\pi + x) = (-1)^n \sin x$, for $0 < x < \frac{\pi}{2}$.

(ii)
$$S = \sin(\pi + x) + \sin(2\pi + x) + \sin(3\pi + x) + \dots + \sin(n\pi + x)$$

$$= -\sin x + \sin x - \sin x + \dots + \sin(n\pi + x)$$

When n is odd $S = -\sin x$

so $-1 < S < 0$, for $0 < x < \frac{\pi}{2}. \quad \checkmark$

When n is even $S = 0$.

So $-1 < S \leq 0. \quad \checkmark$